

Course Code: CSE4011	Course Title Internet of Things	TPC	3	2	4
Version No.	1.0				
Course Pre-requisites/ Co-requisites	None				
Anti-requisites (if any).	None				
Objectives:	1. Students will understand the concepts of Internet of Things and can able to build IoT applications.				
CO's Mapping with PO's and PEO's					
Course Outcomes	Course Outcome Statement			PO's / PEO's	
CO1	Expose state-of-the-art technologies in IoT and its protocols, architectures, and applications.			PO1, PO2, PO3, PO4, PO5, PO6	
CO2	Able to demonstrate Wireless Control of Remote Devices.			PO1, PO2, PO3, PO4	
CO3	Able to analyze various Physical Computing Techniques using Arduino and Raspberry pi.			PO1, PO2, PO3, PO4, PO5, PO6, PO7	
CO4	Able to demonstrate various sensor interfacing using Visual Programming Language			PO1, PO2, PO3, PO4, PO7, PO8	
CO5	Understand judge the applications of SDN, Big Data and cloud			PO1, PO2, PO3, PO4, PO5	
CO6	Able to design and develop smart Application which can interact with Sensors and Actuators.			PO1, PO2, PO3, PO4, PO5, PO6, PO7, PO8	
TOTAL HOURS OF INSTRUCTIONS: 45					
Module No. 1	Introduction to IoT			7 Hours	
Defining IoT, Characteristics of IoT, Sensing, Actuation, Basics of Networking , Communication Protocols					
Module No. 2	Sensor Networks			8 Hours	
Machine to Machine Communication, Difference between IoT and M2M, Software define Networks, Wireless medium access issues, MAC protocol survey, Sensor deployment & Node discovery, Data aggregation & dissemination					
Module No. 3	Introduction to Arduino & Raspberry Pi			8 Hours	
Introduction to Arduino Programming, Types of Arduino, Integration of Sensors and Actuators with Arduino, Integration of IoT with Arduino. Introduction to Raspberry Pi, Implementation of IoT with R-Pi					
Module No. 4	Developing IoTs			7 Hours	
Introduction to Python, Introduction to different IoT tools, Developing applications through IoT tools, Developing sensor based application through embedded system platform, Implementing IoT concepts with python.					
Module No. 5	Introduction to SDN			7 Hours	
Software Defined Networks for IoT, Data Handling analytics, Cloud Computing, Sensor-Cloud, Challenges in IoT, Design challenges, Development challenges, Security challenges, Other challenges.					
Module No. 6	Application of IoT			8 Hours	
Case Study: Home automation, Industry applications, Surveillance applications, Agriculture, Healthcare, Activity Monitoring ,Connectivity Vehicles : Smart Grids & Other IoT applications..					
Text Books					
1. "The Internet of Things: Enabling Technologies, Platforms, and Use Cases", by Pethuru Raj and Anupama C. Raman (CRC Press) 2017.					

References

1. Arshdeep Bahga and Vijay Madisetti "Internet of Things: A Hands-on Approach", 1st Edition, 2014 (Universities Press)
2. Waltenegus Dargie, Christian Poellabauer, "Fundamentals of Wireless Sensor Networks: Theory and Practice", Wiley Publications, 2010

- | | | |
|---|--|-----|
| Mode of Evaluation | Continuous Assessment Tests-60%, Practical Assessment-40% | |
| | Continuous Assessment Test-1 | 20% |
| | Continuous Assessment Test-2 | 20% |
| | Continuous Assessment Test-3 | 20% |
| | Practical Assessment (Mini Project) | 40% |
| Recommended by the Board of Studies on | 2nd BoS 06.07.2018 | |
| Date of Approval by the Academic Council | 2nd Academic Council 21.07.2018 | |

Mode of Evaluation	Continuous Assessment Tests-60%, Practical Assesment-40%	
	Continuous Assessment Test-1	20%
	Continuous Assessment Test-2	20%
	Continuous Assessment Test-3	20%
	Practical Assessment (Mini Project)	40%
Recommended by the Board of Studies on	2nd BoS 06.07.2018	
Date of Approval by the Academic Council	2nd Academic Council 21.07.2018	